Castellvi classification of lumbosacral transitional vertebrae: comparison between conventional radiography, CT, and MRI

Abstract

Background. Reliability and diagnostic accuracy of commonly used diagnostic imaging modalities in the classification of lumbosacral transitional vertebrae (LSTV) are poorly known, and comparative studies are scarce.

Purpose. To compare the diagnostic performance of conventional radiography (CR), CT, and MRI in classifying LSTV.

Materials and Methods. In this retrospective cross-sectional study, a total of 852 patients undergoing lumbar imaging studies in all three modalities were initially assessed for the presence of LSTV using CT scans. 100 patients with LSTV anatomy were identified. An experienced fellowship-trained radiologist performed a gold standard reading using all three modalities. Four readers performed blinded and independent evaluation of these 100 patients on each modality, and inter-reader reliability metrics were analyzed with comparison to gold standard. Statistical software R (4.2.1) was used for the analyses.

Results. We found superior diagnostic efficacy for CT: The sensitivity, specificity, accuracy, and balanced accuracy were 76/93/77/84%, respectively. For MRI, the metrics were 54/88/56/68%, and for CR 32/85/42/59%, respectively. Inter-reader reliability was found to be good for CT (κ = 0.66), and fair for both CR (κ = 0.22) and MRI (κ = 0.38).

Conclusion. CT had the highest diagnostic performance in all measured metrics with a good inter-reader reliability. MRI and CR showed a fairly poor sensitivity and accuracy, and thus consideration should be used when classifying LSTVs with these two modalities.

Key words: Castellvi classification, comparative study, imaging modality, lumbosacral, transitional vertebra

Introduction

Lumbosacral transitional vertebra (LSTV) is a common anatomical variant at the lumbosacral junction of the lumbar spine. The defining feature of LSTV is the enlarged transverse process of the lowest lumbar vertebra which can articulate or fuse with the adjacent sacral ala. LSTVs are usually an incidental finding with debated association with low back pain (LBP) (1). Recent studies with large sample sizes have reported an LSTV prevalence between 11-29% (2-5). The most commonly used LSTV classification by Castellvi in 1984 is based on anteroposterior (AP) radiographs of the lumbar spine. The same study described the Ferguson view, an AP radiograph with 30 degrees of cranial angulation, as the best method for identifying LSTVs (6). It has been suggested that cross-sectional imaging modalities, due to their improved spatial resolution, would be more accurate than conventional radiography (CR) in the classification of LSTVs (7). However, studies comparing various modalities are scarce. One study compared coronal MRI to routine AP radiographs and found AP radiographs to be insufficient in the detection and classification of LSTVs, whereas coronal MRI proved to be highly reliable for the classification of LSTVs (8). In addition, another study found CR to have poor accuracy in the classification of LSTV and the authors proposed coronal plane CT scans as the gold standard for future studies (9). Although it is obvious that modern CT provides superior characterization of bony structures, no studies exist to compare the performance of MRI and CT in the classification of LSTVs. With that in mind, we set out to compare the diagnostic performance of CR, CT, and MRI in the classification of LSTVs.

Materials and methods

Study population and image assessment

Institutional review board approval was obtained and the requirement for informed consent was waived for this retrospective study. We conducted a picture archiving and communication system search between January 1st 2018 and April 30th 2021 for patients who had undergone lumbar imaging studies in all three following modalities: CR, CT, and MRI. We excluded patients with an incomplete view of the lumbosacral junction, a lack of a coronal view on MRI, or spinal implantation at the lumbosacral junction. After the exclusion criteria were applied, the search yielded 852 patients. Using CT scans, 852 patients were assessed for the presence of LSTV using the Castellvi classification. The Castellvi classification divides anatomical changes of the lumbosacral junction into four types (I-IV). In type I, the transverse process of the lowest lumbar vertebra is enlarged and measures at least 19 mm craniocaudally. In type II, the enlarged transverse process forms a pseudoarticulation with the adjacent sacral ala. In type III, the enlarged transverse process forms an osseous fusion with the adjacent sacral ala. Type IV is a combination of types II and III: a pseudoarticulation on one side and an osseous fusion on the contralateral side. Additional subtyping is used for types I-III, where subtype ‘a’ denotes unilateral changes, and subtype ‘b’ denotes bilateral changes (6). In cases of partial osseous fusion of the transverse process and the sacral ala, if more than 50% of the articular surface was fused, it was considered an osseous fusion.

The initial evaluation was carried out by J.H. LSTV anatomy was identified in 100 patients. Four readers - including a first-year radiology resident (J.H.), a fellowship-trained musculoskeletal radiologist with eight years of experience (M.N.), a physiatrist with eight years of experience (J.M.) and an orthopedic spine surgeon with 12 years of experience (P.K.) - performed blinded and

independent evaluation of these 100 cases on each modality (CR, CT, MRI). The gold standard read was carried out by a fellowship-trained musculoskeletal radiologist with more than 20 years of experience (J.N.) assessing the same 100 cases with all the modalities available.

Imaging techniques

On CR, AP weight-bearing view was used; the x-ray beam was focused on L4 vertebra and cropped between the TH12 vertebra and superior border of the coccyx craniocaudally and approximately 1 cm distally from sacroiliac joints laterally. During examination, the subject was asked to hold breath.

On CT, the slice thickness of the scans was 0.6 mm and multiplanar reformatting with coronal and sagittal planes were used. Contrast enhanced abdominal CTs including the entire lumbar spine and lumbosacral junction were collected in the initial PACS search.

On MRI, 94 cases were imaged on 1.5 T and six cases on 3.0 T MRI scanner. A routine clinical lumbar spine protocol including T2-weighted fast spin-echo axial and sagittal planes, T1 fluid-attenuated inversion recovery (FLAIR) sagittal plane as well as short tau inversion recovery (STIR) coronal plane was used. The slice spacing and thickness varied between, 3.5 - 4.4 mm and 3.5 - 4.4 mm in the sagittal, between 0.9 - 4.4 mm and 1.5 - 4.0 mm in the axial, and between 3.6 - 5.5 mm and 3.0 - 5.0 mm in the coronal sequences, respectively.

Statistical analysis

Statistical software R (4.2.1) was used for the analyses. Accuracy, balanced accuracy, sensitivity, specificity, and Cohen's kappa (κ) were calculated to assess inter-reader agreement between the gold standard and the individual reader, and the mean values were calculated from results of all of the four readers for all of the three different imaging modalities, using caret and vld packages. Cohen's kappa values under 0 were considered as no agreement, 0-0.2 as slight, 0.21-0.4 as fair, 0.41-0.6 as moderate, 0.61-0.8 as good, and 0.81-1 as almost perfect agreement.

Results

In total, 100 cases with a mean age of 73 years (range 36 to 93 years) and a gender distribution of 50 males and 50 females were included in this retrospective study. According to the gold standard read comprising the three imaging modalities, 39 cases were identified as Castellvi type I, 31 cases as Castellvi type II, 28 cases as Castellvi type III, and 2 cases as Castellvi type IV. Figure 1 demonstrates an example of Castellvi type IIb LSTV in all three imaging modalities.

The diagnostic performance and inter-reader reliability of each imaging modality are shown in Table 1. As expected, CT yielded the highest sensitivity of 76% and specificity of 94%, whereas CR provided the lowest sensitivity of 32% and specificity of 85% among the modalities. Regarding inter-reader reliability, we found a good agreement for CT ($\kappa = 0.66$), and only a fair agreement for both MRI ($\kappa = 0.38$; 0.24-0.56) and CR ($\kappa = 0.22$). Figure 2 depicts the subtyping results of LSTV among the readers according to each modality, and reader-specific metrics are shown in Table 2.

Discussion

In this comparative study between CR, CT, and MRI to outline the classification of LSTVs, CT was found to have superior diagnostic performance in all metrics with a good inter-reader reliability. MRI showed a surprisingly modest level of sensitivity, and CR remained the poorest modality to classify LSTVs.

In a clinical setting, CT is used if MRI is contraindicated, in preoperative planning, or if the patient has had previous spinal implantation, which can cause problematic imaging artifacts on MRI [10]. CT is also the first-line modality for the imaging of traumatic spinal injuries given its superiority in evaluating osseous anatomy. Nevertheless, CT is limited in the imaging of partially fused articulations between the transverse process and the sacral ala (5, 8). In the modern era of cross-sectional imaging, MRI is the modality of choice when examining patients with low back pain given its high soft tissue contrast within and around the spine (11). Classification of LSTVs and numbering of vertebral bodies is problematic on MRI due to the limited field of view (1), and the use of thick-slice sagittal, axial, and coronal planes. However, the modern isotropic sequences - i.e., 3D MRI - might provide some aid to the evaluation of lumbosacral transitional anatomy. Lastly, standard AP radiographs of the lumbar spine might produce false positive results due to the overlap of the transverse process and the sacrum, while the use of the Ferguson view could decrease this overlap (6). While Ferguson view could show a slightly higher accuracy than standard AP radiographs to detect LSTVs (7, 12), it is not a routinely acquired view in the modern imaging of the lumbar spine (8). CR has its benefits, though, such as easy accessibility, time efficient examinations, low monetary cost, and lesser radiation exposure in comparison to CT (7).

Additionally, CR allows the assessment of the lumbar spine and posture in physiologic weight-bearing conditions as opposed to cross-sectional imaging, which are performed supine.

There are no previous studies to assess the reliability of LSTV classification between MRI and CT. The two previous investigations have only compared CR to either MRI or CT in the detection and classification of LSTVs. A retrospective study of 155 subjects used coronal MRI as the gold standard and found that AP radiographs had an accuracy of 76-84% in detecting LSTVs and an accuracy of 53-58% in classifying LSTVs. The inter-reader reliability in detecting LSTVs was moderate at best ($\kappa = 0.53$) and only fair in classifying LSTVs (weighted $\kappa = 0.39$). In contrast, the inter-reader reliability for coronal MRI was very good for both detection ($\kappa = 0.93$) and classification of LSTVs (weighted $\kappa = 0.83$). The authors suggested that CR should not be used in detection nor classification of LSTVs, and proposed MRI as the gold standard for diagnosing LSTVs (8). In a retrospective study of 91 patients with LSTVs, Hou et al. classified LSTVs on CR and CT; they reported a rather low inter-reader agreement of $\kappa = 0.50$ and $\kappa = 0.49$ (first and second reading, respectively). Additionally, the authors found a misclassification rate of 35% (32/91 patients), all of the misclassified cases were either type II or IV (9). Both of these previous studies are in line with our results, describing the challenges to classify LSTVs especially on CR.

To date, the relationship between LSTVs and LBP remains under debate; however, in the recent literature general lumbosacral degeneration has been linked to LSTVs (5, 13). In large-scale studies utilizing CR, Castellvi type II and IV LSTVs have been suggested to associate with LBP (3, 4). But as established in this study and previous studies (8, 9), CR is not a reliable modality for

accurate subtyping of LSTVs. Therefore, the discrepancies between LSTVs and LBP might be explained to some extent by the used imaging modality.

There are limitations within our study. First, there was a lack of true controls. We did not include any non-LSTV subjects in the final study group and as such, we did not study modality-specific performance in the detection of LSTVs. Second, the quality of radiographs varied greatly. We were not able to apply the Castellvi classification to some studies due to bowel gas and overlapping anatomical structures hindering the visualization of osseous contours of the transverse processes; accordingly, our study cohort represents challenging cases from daily clinical setup. Third, there was no joint pre-training among the readers, but rather each reader was separately trained in the Castellvi classification. Fourth, the gold standard read was performed by a single reader and relied on imaging studies; accordingly, surgical or pathological verification of LSTV anatomy was not possible. Fifth, the imaging examinations of an individual patient were not conducted in a short period of time. With a study period of 40 months, it is possible that pseudoarticulations could have undergone further degenerative changes between imaging examinations and begun to form osseous fusions with the sacral ala affecting the Castellvi classification of our readers.

In conclusion, the results of our study show that CT is the superior imaging modality to classify LSTVs. MRI and CR demonstrate a rather poor diagnostic performance and caution should be used when classifying LSTVs on these modalities.

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Tables

Table 1. Means and ranges of diagnostic performance metrics and inter-reader reliability of each imaging modality. CR = conventional radiography.

	CR	MRI	CT
Accuracy	0.42 (0.34-0.48)	0.56 (0.45-0.7)	0.77 (0.75-0.8)
Balanced accuracy	0.59 (0.57-0.63)	0.68 (0.6-0.76)	0.84 (0.79-0.89)
Sensitivity	0.32 (0.25-0.37)	0.54 (0.34-0.67)	0.76 (0.68-0.85)
Specificity	0.85 (0.84-0.87)	0.88 (0.85-0.91)	0.93 (0.91-0.95)
Cohen's κ	0.22 (0.16-0.32)	0.38 (0.24-0.56)	0.66 (0.63-0.71)

Table 2. Reader-specific diagnostic performance metrics and inter-reader reliability in each imaging modality. The four individual readers are anonymized and represented with the letters A-D. CR = conventional radiography

	CR A	CR B	CR C	CR D	MRI A	MRI B	MRI C	MRI D	CT A	CT B	CT C	CT D
Accuracy	0.34	0.45	0.48	0.40	0.45	0.51	0.70	0.58	0.80	0.77	0.75	0.75
Balanced accuracy	0.57	0.59	0.63	0.58	0.60	0.61	0.76	0.76	0.89	0.82	0.79	0.85
Sensitivity	0.25	0.35	0.37	0.31	0.34	0.39	0.63	0.67	0.85	0.71	0.68	0.79
Specificity	0.84	0.84	0.87	0.84	0.85	0.87	0.91	0.88	0.95	0.94	0.93	0.93
Cohen's κ	0.16	0.21	0.32	0.18	0.24	0.31	0.56	0.39	0.71	0.67	0.63	0.63

Figure Legends

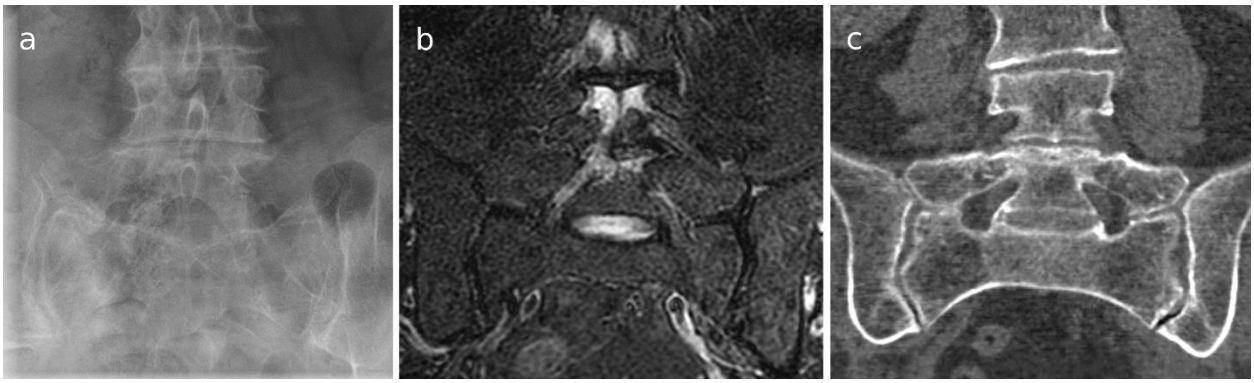
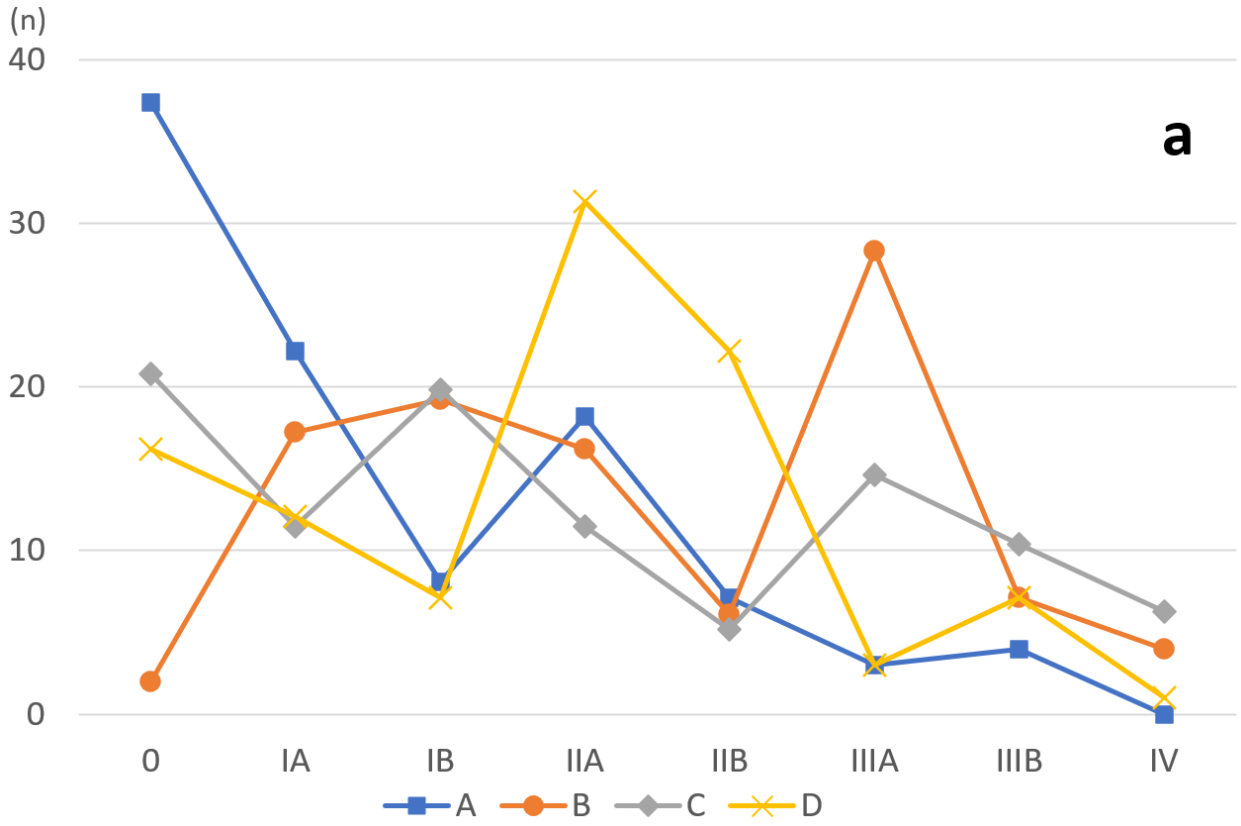
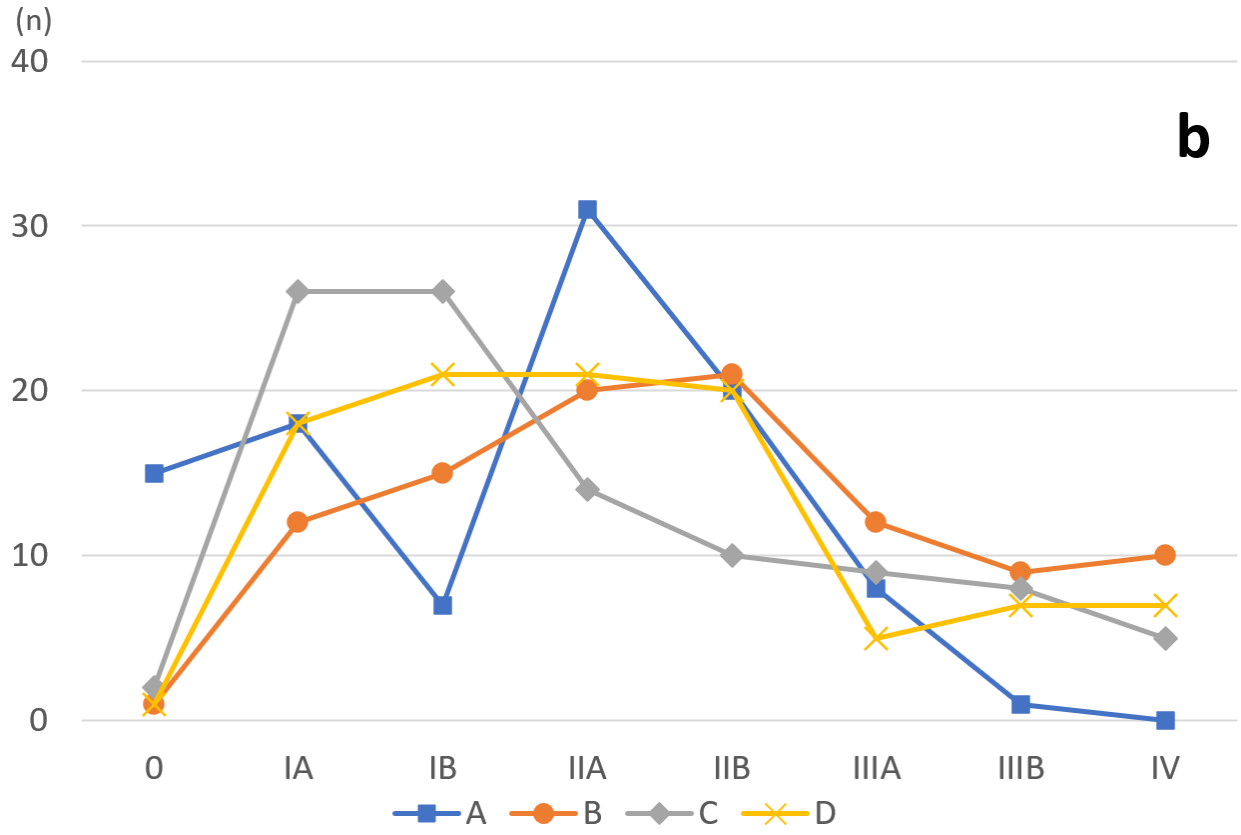


Figure 1. A lumbosacral transitional vertebra (LSTV) of Castellvi type IIb depicted in (a) anteroposterior radiograph, (b) coronal plane short tau inversion recovery (STIR) MR image, and (c) coronal plane CT image.


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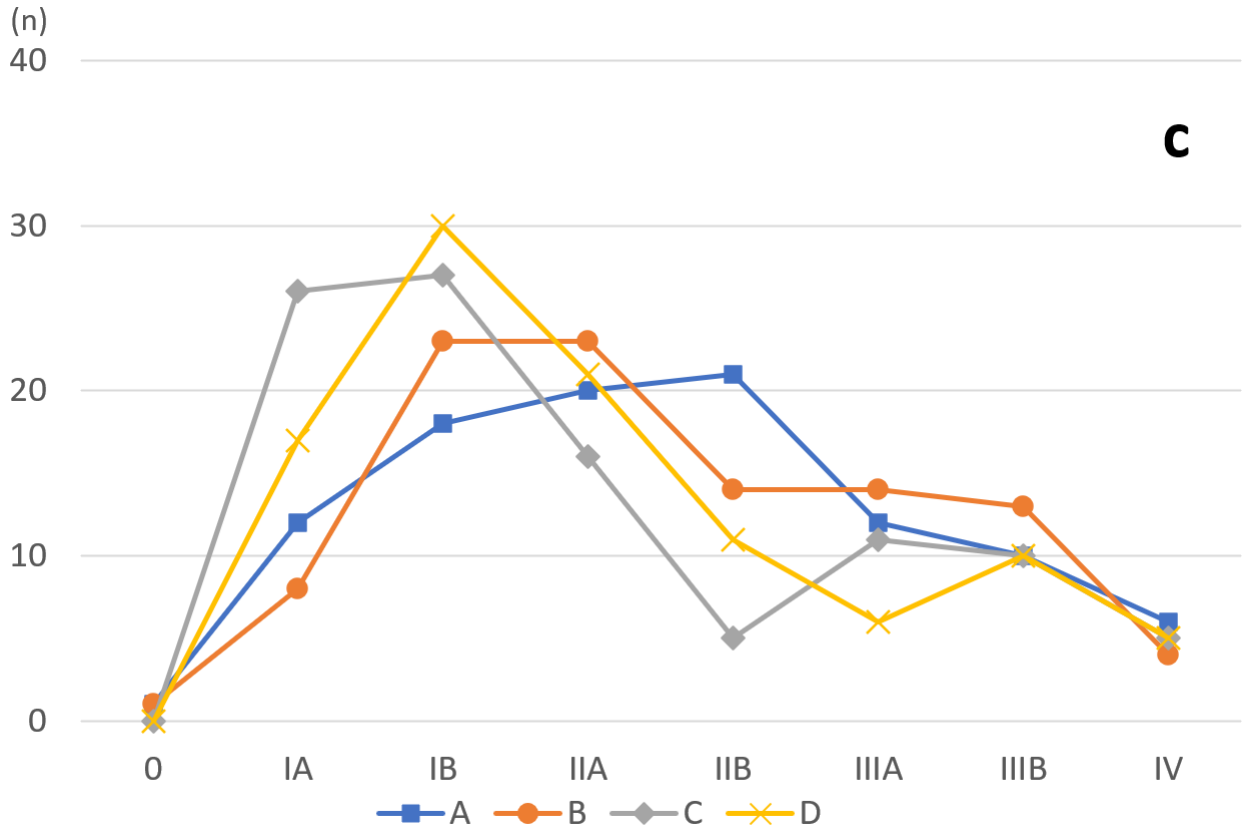


Figure 2. The four readers' lumbosacral vertebra (LSTV) subtyping results are grouped by imaging modalities. The four individual readers are anonymized and represented with the letters A-D. (a) We found CR to have large discrepancies between readers in most Castellvi subtypes, and a high amount of false negative studies. (b) On MRI, the agreement between readers is improved, particularly in types III-IV, and the number of false negatives is considerably reduced. (c) CT demonstrates the least discrepancy and most consistency between readers in the classification.